Do not write outside the box

**0 4** A catalyst is used to increase the rate of decomposition of hydrogen peroxide.

**0 4 . 1** Hydrogen peroxide decomposes to produce water and oxygen.

Write a word equation for the reaction.

**[1 mark]**

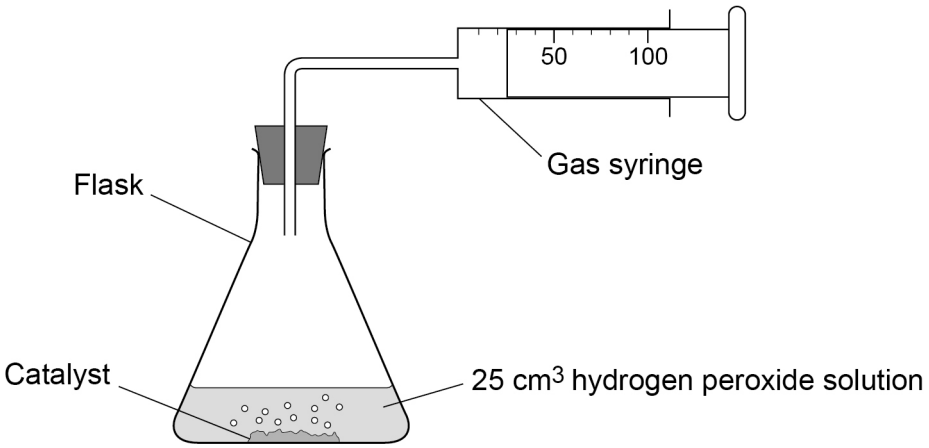
_____ → _____ + _____

A student investigated the rate of decomposition of hydrogen peroxide using three different catalysts:

- manganese dioxide
- copper oxide
- zinc oxide.

**Figure 7** shows the apparatus.

**Figure 7**



Do not write outside the box

This is the method used.

1. Measure 25 $cm^3$ of hydrogen peroxide solution into a flask.
2. Add 0.5 g of manganese dioxide catalyst to the flask.
3. Attach a gas syringe to the flask.
4. Measure the volume of oxygen produced every 20 seconds for 120 seconds.
5. Repeat steps 1 to 4 two more times.
6. Repeat steps 1 to 5 using copper oxide catalyst.
7. Repeat steps 1 to 5 using zinc oxide catalyst.

**0 4 . 2** Which **two** control variables are used in the investigation?

**[2 marks]**

Tick (✓) **two** boxes.

- Mass of catalyst ☐
- Shape of flask ☐
- Time of taking readings ☐
- Volume of hydrogen peroxide solution ☐
- Volume of oxygen produced ☐

**Question 4 continues on the next page**

Do not write outside the box

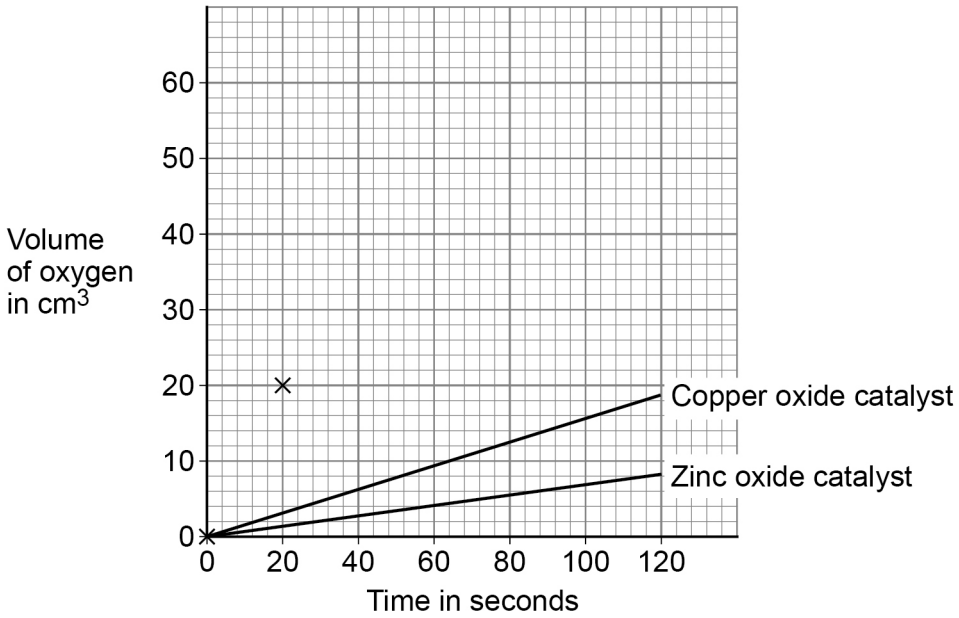
**Table 3** shows the results with manganese dioxide catalyst.

**Table 3**

| **Time in seconds** | 0 | 20 | 40 | 60 | 80 | 100 | 120 |
|---|---|---|---|---|---|---|---|
| **Volume of oxygen in $cm^3$** | 0 | 20 | 32 | 40 | 45 | 49 | 50 |

**Figure 8** shows a graph of the results with copper oxide catalyst and with zinc oxide catalyst.

**Figure 8**




**0 4 . 3** Complete **Figure 8**.

You should:

- plot the data for the manganese dioxide catalyst from **Table 3**
- draw a line of best fit.

The first two points have been plotted for you.

**[3 marks]**

Do not write outside the box

**0 4 . 4** Which is the best catalyst?

Give **one** reason for your answer.

Use the completed **Figure 8**.

**[2 marks]**

Tick (✓) **one** box.

Copper oxide ☐

Manganese dioxide ☐

Zinc oxide ☐

Reason _____

_____

**0 4 . 5** The student repeated the investigation with manganese dioxide catalyst.

The results were all close to the true value.

Complete the sentence.

Choose the answer from the box.

**[1 mark]**

| | | |
|---|---|---|
| **accurate** | **approximate** | **random** |

Results that are close to the true value are _____.

**Question 4 continues on the next page**

Do not write outside the box

**0 4 . 6** The student repeated the investigation using manganese dioxide catalyst at a higher temperature.

Complete the sentence.

Choose the answer from the box.

**[1 mark]**

| | | |
|---|---|---|
| **decrease** | **stay the same** | **increase** |

At a higher temperature, the rate of reaction will _____.

10